

Q1)

a) Data type of all columns in the "customers" table.

Ans:

```
SELECT column_name, data_type
FROM `spartan-avenue-414916.Target.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = "customers";
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS
Row	column_name	data_type			
1	customer_id	STRING			
2	customer_unique_id	STRING			
3	customer_zip_code_prefix	INT64			
4	customer_city	STRING			
5	customer_state	STRING			

Insights:

1. Above table is giving data types for all column related to customers table
2. String data type for the customer_id, customer_unique_id, customer_city and customer_state

Recommendation: NA(Not Applicable)

b) Get the time range between which the orders were placed.

Ans:

```
SELECT
    min(order_purchase_timestamp) AS `min_timestamp`,
    max(order_purchase_timestamp) AS `max_timestamp`
FROM
    `spartan-avenue-414916.Target.orders`
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	E
Row	min_timestamp	max_timestamp			
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC			

Insights:

1. The operations of target company in Brazil has orders between the 2016 and 2018
2. We can fetch the data by observing the above table First order placed on 2016-09-04 and Last order 2018-10-17
3. Orders table is giving us the

Recommendation: NA

c) Count the Cities & States of customers who ordered during the given period.

Ans:

```
SELECT
    COUNT(DISTINCT c.customer_city) AS `count_of_city`,
    COUNT(DISTINCT c.customer_state) AS `count_of_state`,
    min(o.order_purchase_timestamp) AS `min_timestamp`,
    max(o.order_purchase_timestamp) AS `max_timestamp`
FROM
    `spartan-avenue-414916.Target.customers` AS `c` INNER JOIN `spartan-avenue-414916.Target.orders` AS `o` ON c.customer_id=o.customer_id
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	count_of_city	count_of_state	min_timestamp	max_timestamp		
1	4119	27	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC		

Insights:

1. The orders from 2016 to 2018 has covered most regions observed good response
2. Most of the geographical distribution area has covered with minimal time
3. The city or state which is having low order rate and high order rate will get good understanding of expand the business in respective cities or states

Recommendation:

1. We can increase the supply chain so that productivity and time management
2. We need to maintain more inventory so that we cant ran out of inventories in some cities or states where order rate is high
3. We need to make advertisements and promotions to attaract customers in cities where order rate is low

Q2)

a) Is there a growing trend in the no. of orders placed over the past years?

Ans:

```
SELECT
    EXTRACT(YEAR FROM order_purchase_timestamp) AS `yearwise_orders`,
    COUNT(order_id) AS `total_orders`
FROM
    `spartan-avenue-414916.Target.orders`
GROUP BY 1
ORDER BY
    yearwise_orders
```

Screenshot:

JOB INFORMATION		RESULTS	CHART
Row	yearwise_orders	total_orders	
1	2016	329	
2	2017	45101	
3	2018	54011	

Insights:

1. The above table will be giving us growing trend in the over past years
2. Target is doing a good job over the past years if we analyze year-on-year table as mentioned above
3. Supply chain, targeted marketing, business strategies, customer engagement helps Target to reach the growing trend on y-o-y

Recommendation:

1. To improve in the marketing and become a well-known brand in Brazil, Target has to maintain more resources to maintain the supply chain and logistics operations
2. Growing trend observed over the past years but still some of the cities and states have a lower order rate, tie up with local retailers and do promotions as much as possible

b) Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Ans:

```
SELECT
    FORMAT_DATE('%Y-%m', order_purchase_timestamp) AS `year_with_month`,
```

```

COUNT(*) AS `total_orders`
FROM
`spartan-avenue-414916.Target.orders`
GROUP BY
FORMAT_DATE( '%Y-%m',order_purchase_timestamp)
ORDER BY
year_with_month ASC;

```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON
Row	year_with_month	total_orders		
1	2016-09	4		
2	2016-10	324		
3	2016-12	1		
4	2017-01	800		
5	2017-02	1780		
6	2017-03	2682		
7	2017-04	2404		
8	2017-05	3700		
9	2017-06	3245		
10	2017-07	4026		

Insights:

1. There are high peaks observed between 2017-Nov to 2018-Aug , where most of the customers ordered
2. There are lower peaks 2016-Sept to 2017-Jan and 2018-Sept to 2018-Nov
3. There is sudden increase in orders from 2017-Jan to 2017-Feb
4. There is sudden drop in orders from 2018-Aug to 2018-Sept

Recommendation:

1. When high order rate is observed then we need to maintain inventory and business supply chain
2. When lower orders is observed then more promotion, more advertisements, discount need to be offered so that customer attracting towards the offers
3. Continuously monitor and analyse the month-on-month data to understand the trend and continuous improvement towards deliveries for the customers

c) During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- **0-6 hrs : Dawn**
- **7-12 hrs : Mornings**
- **13-18 hrs : Afternoon**
- **19-23 hrs : Night**

Ans: `SELECT`

```

(
CASE
WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN
"Dawn"
WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN
"Mornings"
WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN
"Afternoon"
ELSE "Night"
END
) AS `time_of_the_day`,
COUNT(*) AS `total_orders`
FROM
`spartan-avenue-414916.Target.orders`
GROUP BY 1
ORDER BY
time_of_the_day ASC;

```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON
Row	time_of_the_day	total_orders		
1	Afternoon	38135		
2	Dawn	5242		
3	Mornings	27733		
4	Night	28331		

Insights:

- 1.highest orders placed in afternoon i.e between 13 to 18 hours
- 2.lowest orders placed in Dawn i.e between 0 to 6 hours
- 3.same level of orders between Morning and Night i.e 7 to 12 hours and 19 to 23 hours

Recommendation:

- 1.order rate during afternoon is more compare to morning and night to increase order rate in morning and night hours by sending the personalized offers and discounts in festive seasons
- 2.To increase order rate where order rate is less in odd hours increase in staffing levels and intime delivery
3. Anticipate demand patterns associated with different times of the day and adjust product availability accordingly.

Q3)

a) Get the month on month no. of orders placed in each state.

Ans:

```
SELECT DISTINCT
  EXTRACT(YEAR FROM order_purchase_timestamp) AS `yearwise_orders`,
  EXTRACT(MONTH FROM o.order_purchase_timestamp) AS `monthwise_orders`,
  c.customer_state,
  COUNT(*) AS `orders_by_statewise`
FROM
  `spartan-avenue-414916.Target.orders` AS `o` INNER JOIN `spartan-avenue-414916.Target.customers` AS `c`
  ON o.customer_id=c.customer_id
GROUP BY 1,2,3
ORDER BY
  yearwise_orders,monthwise_orders
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	E
Row	yearwise_orders	monthwise_orders	customer_state	orders_by_statewise		
1	2016	9	RR			1
2	2016	9	RS			1
3	2016	9	SP			2
4	2016	10	SP			113
5	2016	10	RS			24
6	2016	10	RJ			56
7	2016	10	MT			3
8	2016	10	GO			9
9	2016	10	MG			40
10	2016	10	CE			8

Insights:

1. The above table has orders by each state
2. Table which also has month-on-month analysis of orders
3. The most orders has observed by SP state in top 10 results

Recommendation:

1. Orders in 2016 are less order rate in growing trend whereas 2017 has good order rate compare to last year, continuous monitoring is helpful for improving the business
2. Warehouse in cities need to be increased to save the delivery time and customer waiting time
3. Improvement in logistics and resource need to be hired in festive seasons to avoid delay in customer order delivery

b) How are the customers distributed across all the states?

Ans:

```
SELECT
  COUNT(DISTINCT customer_id) AS `total_customers`,
  customer_state
FROM
  `spartan-avenue-414916.Target.customers`
GROUP BY
  customer_state
ORDER BY
```

total_customers DESC

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSC
Row	total_customers	customer_state		
1	41746	SP		
2	12852	RJ		
3	11635	MG		
4	5466	RS		
5	5045	PR		
6	3637	SC		
7	3380	BA		
8	2140	DF		
9	2033	ES		
10	2020	GO		

Insights:

1. Highest number of customers are from state "SP"
2. Lowest number of customers are from state "GO" and "ES"
3. Similar no of customers is from state "RS","PR"

Recommendation:

1. To increase the no of customers improve the customer service and engage the customers with attractive offers and discounts
2. Increase the operational efficiency and product availability.

Q4)

a) **Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).**

You can use the "payment_value" column in the payments table to get the cost of orders.

Ans:

```
WITH orders_2017 AS (
SELECT
    ROUND(SUM(p.payment_value),2) AS `total_payment_2017`
FROM
    `spartan-avenue-414916.Target.payments` AS `p` INNER JOIN `spartan-avenue-414916.Target.orders` AS `o`
    ON p.order_id=o.order_id
WHERE
    EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
),
orders_2018 AS (
SELECT
    ROUND(SUM(p.payment_value),2) AS total_payment_2018
FROM
```

```

`spartan-avenue-414916.Target.payments` AS `p` INNER JOIN `spartan-
avenue-414916.Target.orders` AS `o`
ON p.order_id=o.order_id
WHERE
EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
)
SELECT
ROUND(((orders_2018.total_payment_2018 - orders_2017.total_payment_2017)
/ orders_2017.total_payment_2017 * 100),2) AS percentage_increase
FROM
orders_2017,
orders_2018;

```

Screenshot:

JOB INFORMATION		RESULTS
Row	percentage_increase	
1	136.98	

Insights:

1. Average orders between 2017 and 2018 are counted (Including Jan to Aug only).
2. To understand the percentage increase from 2017 to 2018 is we need to get payment value for each order
3. Cost of orders we get from payment value from payment table.
4. Finally we will get the percentage increase in cost of orders from 2017 to 2018

Recommendation:

1. Sudden fall in increase in orders 2018 Sept to 2018 Oct therefore fall in cost of orders if we calculate from 2017 Sept to 2017 Dec
2. To increase orders from lower order rate maintain customer service and logistics operations ease and make more promotions and advertisements

b) Calculate the Total & Average value of order price for each state.

Ans:

```

SELECT DISTINCT
ROUND(AVG(oi.price),2) AS `avg_price`,
ROUND(SUM(oi.price),2) AS `total_price`,
c.customer_state
FROM
`spartan-avenue-414916.Target.order_items` AS `oi` INNER JOIN
`spartan-avenue-414916.Target.orders` AS `o`
ON oi.order_id=o.order_id
INNER JOIN `spartan-avenue-414916.Target.customers` AS `c`
ON o.customer_id=c.customer_id

```



```

GROUP BY
    c.customer_state
ORDER BY
    c.customer_state

```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION
Row	avg_price	total_price	customer_state		
1	173.73	15982.95	AC		
2	180.89	80314.81	AL		
3	135.5	22356.84	AM		
4	164.32	13474.3	AP		
5	134.6	511349.99	BA		
6	153.76	227254.71	CE		
7	125.77	302603.94	DF		
8	121.91	275037.31	ES		
9	126.27	294591.95	GO		
10	145.2	119648.22	MA		

Insights:

1. Total and average price per order for each state
2. avg_price will give us average price for total orders done by each state
3. total_price will give us sum price for total orders done by each state

Recommendation: NA

a) Calculate the Total & Average value of order freight for each state.

Ans:

```

SELECT DISTINCT
    ROUND(AVG(oi.freight_value),2) AS `avg_freight_value`,
    ROUND(SUM(oi.freight_value),2) AS `total_frieght_value`,
    c.customer_state
FROM
    `spartan-avenue-414916.Target.order_items` AS `oi` INNER JOIN `spartan-
avenue-414916.Target.orders` AS `o`
    ON oi.order_id=o.order_id
    INNER JOIN `spartan-avenue-414916.Target.customers` AS `c`
    ON o.customer_id=c.customer_id
GROUP BY
    c.customer_state
ORDER BY
    c.customer_state

```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION
Row	avg_freight_value	total_freight_value	customer_state		
1	40.07	3686.75	AC		
2	35.84	15914.59	AL		
3	33.21	5478.89	AM		
4	34.01	2788.5	AP		
5	26.36	100156.68	BA		
6	32.71	48351.59	CE		
7	21.04	50625.5	DF		
8	22.06	49764.6	ES		
9	22.77	53114.98	GO		
10	38.26	31523.77	MA		

Insights:

1. This table will give us average freight value and total freight value for each state
2. Avg_freight_value will give average freight value for each state
3. Total_freight_value will give total freight value for each state

Recommendation: NA

Q5)

- a) Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- $\text{time_to_deliver} = \text{order_delivered_customer_date} - \text{order_purchase_timestamp}$
- $\text{diff_estimated_delivery} = \text{order_delivered_customer_date} - \text{order_estimated_delivery_date}$

Ans:

```
SELECT
    (EXTRACT(DAY FROM order_delivered_customer_date) - EXTRACT(DAY FROM
order_purchase_timestamp)) AS `time_to_deliver`,
    (EXTRACT(DAY FROM order_delivered_customer_date) - EXTRACT(DAY FROM
order_estimated_delivery_date)) AS `diff_estimated_delivery`
FROM
    `spartan-avenue-414916.Target.orders`
```

Screenshot:

JOB INFORMATION		RESULTS	CHART
Row	time_to_deliver	diff_estimated_deliver	
1	0	1	
2	0	-2	
3	-28	-11	
4	-29	-12	
5	21	12	
6	-27	-13	
7	-29	-12	
8	23	17	
9	-27	-13	
10	25	24	

Insights:

- 1.table has delivered time for each order from the order purchase date which is time_to_deliver
- 2.It will give the difference between delivered date and estimated delivery date.

Recommendation:

- 1.By observing the table we can improve the delivery time by keeping more resources so that customer service will not be affected
- 2.we can improve performance of logistics, supply chain and maintaining of inventory

b) Find out the top 5 states with the highest & lowest average freight value.

Ans: `WITH `CTE` AS
(
SELECT
ROUND(AVG(oi.freight_value),2) AS `avg_freight_value`,
c.customer_state
FROM
`spartan-avenue-414916.Target.order_items` AS `oi` INNER JOIN `spartan-avenue-414916.Target.orders` AS `o`
ON oi.order_id=o.order_id
INNER JOIN `spartan-avenue-414916.Target.customers` AS `c` ON
o.customer_id=c.customer_id
GROUP BY
c.customer_state
)
(
SELECT
customer_state,
avg_freight_value
FROM
CTE
ORDER BY
avg_freight_value DESC
LIMIT 5
)`

```

UNION ALL
(
SELECT
    customer_state,
    avg_freight_value
FROM
    CTE
ORDER BY
    avg_freight_value ASC
LIMIT 5
)

```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON
Row	customer_state	avg_freight_value		
1	RR	42.98		
2	PB	42.72		
3	RO	41.07		
4	AC	40.07		
5	PI	39.15		
6	SP	15.15		
7	PR	20.53		
8	MG	20.63		
9	RJ	20.96		
10	DF	21.04		

Insights:

1. Highest average freight value in above table is state "RR"
2. Lowest average freight value is state "SP"

Recommendation: NA

c) Find out the top 5 states with the highest & lowest average delivery time.

Ans:

```
WITH `CTE` AS
(
  SELECT
    ROUND(AVG(EXTRACT(DAY FROM order_delivered_customer_date) -EXTRACT(DAY FROM
order_purchase_timestamp)),2) AS `avg_deliver_time`,
    c.customer_state
  FROM
    `spartan-avenue-414916.Target.order_items` AS `oi` INNER JOIN `spartan-avenue-
414916.Target.orders` AS `o`
    ON oi.order_id=o.order_id
    INNER JOIN `spartan-avenue-414916.Target.customers` AS `c` ON
    o.customer_id=c.customer_id
  GROUP BY
    c.customer_state
)

(SELECT
  customer_state,
  avg_deliver_time
FROM
  CTE
ORDER BY
  avg_deliver_time DESC
LIMIT 5)

UNION ALL

(SELECT
  customer_state,
  avg_deliver_time
FROM
  CTE
ORDER BY
  avg_deliver_time ASC
LIMIT 5)
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON
Row	customer_state	avg_deliver_time		
1	AP	3.77		
2	RR	2.37		
3	PB	1.41		
4	PE	1.33		
5	PI	1.23		
6	AM	-1.52		
7	AC	-1.27		
8	RO	-0.11		
9	AL	0.04		
10	MA	0.12		

Insights:

1. The highest average delivery time is from state "AP"
2. Lowest average delivery time is from state "AM"

Recommendation:

1. The average delivery time for some states taking longer than usual to improve the delivery time continuous improvement in logistics
2. more resources faster customer service and support
3. analyze the data on month on month to get better understanding on customers trending behaviors in orders

d) Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

Ans:

```
SELECT
AVG(DATE_DIFF(o.order_delivered_customer_date,o.order_estimated_delivery_date,DAY))
`diff_estimated`,
c.customer_state
`diff_estimated`,
c.customer_state
FROM
`spartan-avenue-414916.Target.orders` AS `o` INNER JOIN `spartan-avenue-
414916.Target.customers` AS `c`
ON o.customer_id=c.customer_id
GROUP BY
c.customer_state
ORDER BY
diff_estimated ASC
LIMIT 5
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSO
Row	diff_estimated	customer_state		
1	-19.7625000000...	AC		
2	-19.1316872427...	RO		
3	-18.7313432835...	AP		
4	-18.6068965517...	AM		
5	-16.4146341463...	RR		

Insights:

1. Top 5 states who received the delivery much faster as per given estimated delivery time
2. To get the top 5 states we need to calculate the difference between average of actual and estimated delivery time

Recommendation:

1. We also get to know the in which state we are lacking in delivery the order on time as per estimated delivery time ,we can improve the delivery intime by hiring more resources and expanding warehouses in more orders areas

Q6)

a) Find the month on month no. of orders placed using different payment types.

Ans: `SELECT DISTINCT`

```
EXTRACT(YEAR FROM o.order_purchase_timestamp) AS `year_on_year`,
EXTRACT(MONTH FROM o.order_purchase_timestamp) AS `month_on_month`,
p.payment_type,
COUNT(*) AS `no_of_orders`
FROM
`spartan-avenue-414916.Target.orders` AS `o` INNER JOIN `spartan-avenue-414916.Target.payments` AS `p`
ON o.order_id=p.order_id
GROUP BY
1,2,3
ORDER BY
year_on_year ASC,month_on_month ASC
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	E
Row	year_on_year	month_on_month	payment_type	no_of_orders		
1	2016	9	credit_card	3		
2	2016	10	credit_card	254		
3	2016	10	UPI	63		
4	2016	10	voucher	23		
5	2016	10	debit_card	2		
6	2016	12	credit_card	1		
7	2017	1	credit_card	583		
8	2017	1	UPI	197		
9	2017	1	voucher	61		
10	2017	1	debit_card	9		

Insights:

1. The table has data of year-on-year, month-on-month analysis ,payment type and number of orders for each payment type

Recommendation:

1. After analyzing the data we know that most of the payments done by credit cards as we on basis of month-on-month ,we can increase the credit cards by offering some credit points for every payments
2. Data also gives increasing in UPI payments like credit cards ,UPI mode of payment is cashless and on of the secure payment modes we can improve the website operations while paying via credit cards ,UPI ,debit cards so that customers more attractive towards cashless payments

b) Find the no. of orders placed on the basis of the payment installments that have been paid.

Ans: `SELECT`

```
payment_installments,  
COUNT(DISTINCT o.order_id) AS `no_of_orders`  
FROM  
`spartan-avenue-414916.Target.orders` AS `o` INNER JOIN `spartan-avenue-414916.Target.payments` AS `p`  
ON o.order_id=p.order_id  
WHERE  
payment_installments=1  
GROUP BY  
payment_installments
```

Screenshot:

JOB INFORMATION		RESULTS	CHART
Row	payment_installment	no_of_orders	
1	payment_installments	49060	

Insights:

1. Table will give us the how many transactions done via installments with full payment paid
2. We can observe the tables as payment installments and number of orders

Recommendation:

1. payment via installments has been good mode of payments for customers who are often using the payment_installment option
2. improve in the installment option is increase in more customers choosing the payment mode which will give cashless mode
3. To increase the orders via installments , we can provide offers and discounts for paying via installments , customers need to have understanding of secure payments while choosing the payment_installments